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# Beyond the Scale in Indian Carnatic Music

*Varying Emotions Across Ragas of the Same Scale*

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MUSIC WORKSHOP • SEMESTER PROJECT PROPOSAL

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## BACKGROUND AND RATIONALE

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A raga is not its scale. Two ragas can draw on the same set of swaras, the same arohana and avarohana skeleton, and still stand apart as distinct musical entities, each with its own characteristic phrases, gamakas, emphasised notes, and settled emotional character. This is a foundational claim of the Carnatic system, familiar to any trained listener and encoded in the distinction between a melakarta and its janya ragas. The identity of a raga rests on how those notes are approached, ornamented, and joined, rather than the scale alone.

The empirical literature on raga and emotion has not tested this claim directly. Existing work establishes that distinct ragas evoke distinct emotions, and that even culturally unfamiliar listeners can identify the intended mood [1, 2, 3]. It also shows that the same raga rendered at different tempo shifts in emotional character [3, 4]. But this work typically studies ragas of different scales, and repeatedly treats the shared-scale case as a known fact rather than a measured one. Mathur et al. note in passing that ragas can share the same notes yet keep separate identities, and then study unrelated ragas [3]. The assumption that phrase grammar, not scale, carries a raga's affect is asserted across the field but has not been shown to hold for listeners. Carnatic evidence is especially thin, and tends to treat a whole raga as a single fixed emotional object [5, 6, 7, 8].

This project tests the assumption directly. By holding the scale constant and varying only the raga, it asks whether listeners perceive an emotional difference that the scale alone cannot explain. If they do, then emotion in Carnatic music cannot be reduced to the notes of the scale, and the raga's identity is shown to live in its treatment of those notes rather than in their bare selection.

## RESEARCH QUESTIONS

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**RQ1** (*primary*). Do janya ragas of a single melakarta, sharing the same scale, evoke distinct emotional responses in listeners? A confirmed difference would show that emotion in Carnatic music cannot be attributed to the scale alone, and that the raga's affective identity rests on its characteristic treatment of the notes.

**RQ2** (*secondary, exploratory*) Within a single raga, do different characteristic phrases evoke different emotional responses? A confirmed difference would suggest that affect is nuanced even below the level of the raga: while a composition or a rendition can paint an overall mood, that mood cannot be pinned to the scale, nor fully to the raga itself.

## METHOD

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The study runs as a short listener survey built on excerpts drawn from existing recordings. It requires no instruments or laboratory measurement.

- **Stimuli.** For RQ1, janya ragas of a shared melakarta. The primary set draws on janyas of Kharaharapriya (candidates: Kanada, Reethigowla, Abheri); a secondary set draws on janyas of Shankarabharanam (candidates: Nalinakanthi, Kadanakuthuhalam). Ragas are not yet finalised. All clips are purely instrumental and are matched as far as possible on timbre, pitch, and speed, so that any perceived difference is attributable to the raga rather than to surface features. Sources are existing recordings (including publicly available material).
- **Familiarisation segment.** Each clip is preceded by a short, plain listening segment in the same timbre, so that listeners are settled into the sound before rating and are not biased by the novelty of a new tone.
- **Expert vetting.** Trained artists review the clips before release, with the intent of the survey made explicit to them, to confirm that each excerpt is idiomatic and that the matched-surface constraint holds.
- **Measures.** For each excerpt: valence and arousal (a five-point or pictorial SAM scale) and one free-response term for the perceived emotion. Clip order is randomised to control for ordering effects.
- **Participants.** The primary Kharaharapriya set targets approximately 100 responses; the secondary Shankarabharanam set approximately 30–50. Listeners are drawn from the general student body. A single item records prior Carnatic training, allowing comparison between trained and untrained listeners. The survey is designed to take five to eight minutes.
- **Analysis.** Descriptive statistics and the distribution of ratings across ragas of the shared scale, reported with effect sizes rather than significance tests alone. No causal claims are made about specific musical features.

For RQ2, where pursued, the same raga is presented in two different characteristic phrases without identifying it as such, and emotional responses are compared across the two.

## METHODOLOGICAL SAFEGUARDS

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- **Matched surface features.** Instrumental clips matched on timbre, pitch, and speed isolate the raga as the variable of interest and remove the most obvious confounds.
- **Expert vetting of stimuli.** Independent review by trained artists guards against unidiomatic or poorly matched excerpts, and keeps the stimulus selection from resting on the author's judgement alone.
- **Familiarisation segments.** Settling listeners into the timbre before each rating reduces novelty bias.
- **Validated scales.** An established valence/arousal or SAM scale improves comparability at no additional cost.
- **Stated scope.** The design is cross-sectional and correlational, and does not attempt to identify which specific feature carries the perceived difference, since these covary in natural renditions. Where the discussion touches on this, it is treated as interpretation, not measurement.

## EXPECTED CONTRIBUTION AND OUTPUTS

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The project tests a foundational claim about Carnatic music that the literature assumes but has not demonstrated: that emotion is not reducible to the scale, and that a raga's affective identity lives in its treatment of the notes. A confirmed within-melakarta difference in Carnatic music, on matched instrumental stimuli, would be a direct and novel demonstration of this. The secondary question, if pursued, extends the point below the level of the raga.

The survey, together with the full set of vetted clips, will be released as an open-source artifact and hosted publicly, so that the stimuli and instrument can be reused and scrutinised. Should RQ2 prove hard to include in the current scope, the project stands on RQ1 alone.

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## REFERENCES

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- [1] Balkwill, L., & Thompson, W. F. (1999). A cross-cultural investigation of the perception of emotion in music. *Music Perception*, 17(1), 43–64.
- [2] Chordia, P., & Rae, A. (2008). Understanding emotion in raag: an empirical study of listener responses. *Proc. CMMR*.
- [3] Mathur, A., et al. (2015). Emotional responses to Hindustani raga music: the role of musical structure. *Frontiers in Psychology*, 6:513.
- [4] Velankar, M., et al. (2017). Study of emotion perception for Indian classical raga music. *Proc. SysMus17*, QMUL.
- [5] Ravi, M., & Govindaraj, M. S. (2021). Emotional impact of four specific ragas of Carnatic music. *Intl. J. Alternative and Complementary Medicine*.
- [6] (2025). Impact of integrated Indian Classical Carnatic ragas on Positive and Negative Affect. *Intl. J. Interdisciplinary Approaches in Psychology*, 3(3).
- [7] Koduri, G. K., & Indurkha, B. (2010). A behavioral study of emotions in South Indian classical music. *SAPMIA '10*, ACM.
- [8] Priyadarshini, A., et al. RaagaDhvani: an augmented multi-feature dataset for emotion recognition in Carnatic music.
- [9] Ganguli, K. K., & Rao, P. (2018). On the distributional representation of ragas: experiments with allied raga pairs. *TISMIR*, 1(1).
- [10] Mathur, A., et al. (2019). Cultural differences in the use of acoustic cues for musical emotion experience. *PLOS ONE*, 14(9).
- [11] Music and emotion: a case for North Indian classical music. (2017). *Frontiers in Psychology*, 8:2115.
- [12] Meyer, L. B. (1956). *Emotion and Meaning in Music*. Univ. of Chicago Press.
- [13] Huron, D. (2006). *Sweet Anticipation: Music and the Psychology of Expectation*. MIT Press.